

PART I - FOUNDATION PLC PROJECTS

Project 4 - Temperature Control System

Version 1 - Foundation PLC Project

PLC Platform: Allen-Bradley MicroLogix 1100

Programming Software: RSLogix Micro Starter Lite 8.30

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1. Project Overview

The Temperature Control System was developed to introduce process control using PLC comparison instructions and timer-based alarm monitoring. The system uses a simulated temperature value stored in integer register N7:0 and automatically controls heating, cooling, and high-temperature alarm outputs according to predefined thresholds.

When the temperature is below 70°F, Heater O:2/0 is energized. When the temperature rises above 80°F, Cooling Fan O:2/1 is energized. Between these two limits, both outputs remain OFF, creating a normal operating deadband.

I also added a high-temperature monitoring function. When the temperature remains above 90°F continuously until Timer T4:0 reaches its 10-second preset, Alarm O:2/2 energizes.

This project expanded my PLC foundation from discrete sensor-based control into value-based decision making using integer data, comparison instructions, and delayed alarm logic.

2. Engineering Objective

The objective of this project was to develop and verify a PLC program capable of automatically controlling heating and cooling outputs based on a simulated temperature value while also providing a delayed indication for a sustained high-temperature condition.

- Turn the Heater ON when the temperature is below 70°F.
- Turn the Heater OFF when the temperature reaches 70°F or higher.
- Keep both Heater and Cooling Fan OFF between 70°F and 80°F.
- Turn the Cooling Fan ON when the temperature exceeds 80°F.
- Monitor temperatures above 90°F using a TON instruction.
- Energize the Alarm after the high-temperature condition remains present through the 10-second timer period.

This project gave me practical experience with LES, GRT, TON, XIC, OTE, integer registers, and timer status bits.

3. Functional Requirements

- N7:0 shall represent the simulated process temperature.
- Heater O:2/0 shall energize when $N7:0 < 70$.
- Heater O:2/0 shall remain OFF when $N7:0 \geq 70$.
- Cooling Fan O:2/1 shall energize when $N7:0 > 80$.
- Cooling Fan O:2/1 shall remain OFF when $N7:0 \leq 80$.
- Both Heater and Cooling Fan shall remain OFF from 70°F through 80°F.
- Timer T4:0 shall be enabled when $N7:0 > 90$.
- T4:0 shall use a 0.01-second time base and preset of 1000.
- The resulting high-temperature delay shall be 10 seconds.
- Alarm O:2/2 shall remain OFF while T4:0/DN is false.
- Alarm O:2/2 shall energize when T4:0/DN becomes true.
- If temperature returns to 90°F or below before the timer completes, the non-retentive timer shall reset.

- When T4:0 resets, its Done bit shall clear and the Alarm shall turn OFF.

4. System Components

Component	Identification	Function
PLC Platform	Allen-Bradley MicroLogix 1100	Executes the temperature-control program
Temperature Value	N7:0	Stores the simulated temperature
Heater	O:2/0	Heating command
Cooling Fan	O:2/1	Cooling command
High-Temperature Timer	T4:0	Monitors sustained temperature above 90°F
Alarm	O:2/2	Provides high-temperature indication
Programming Software	RSLogix Micro Starter Lite 8.30	Program development and monitoring

In this Version 1 project, N7:0 is a simulated integer temperature value. The ladder logic does not contain analog input scaling, engineering-unit conversion, or temperature-sensor diagnostics.

5. I/O and Data Assignment

Output Assignment

Device	PLC Address	Type	Description
Heater	O:2/0	Digital Output	Heating command
Cooling Fan	O:2/1	Digital Output	Cooling command
Alarm	O:2/2	Digital Output	High-temperature indication

Internal PLC Data

Function	Address	Type	Purpose
Simulated Temperature	N7:0	Integer	Represents process temperature
High-Temperature Timer	T4:0	TON	Monitors sustained temperature above 90°F
Timer Done	T4:0/DN	Status Bit	Enables alarm after timing completes
Timer Time Base	-	0.01 second	Timer resolution
Timer Preset	-	1000	High-temperature delay setting

Timer delay: $1000 \times 0.01 \text{ second} = 10.00 \text{ seconds}$.

6. Control Strategy

I divided the control strategy into three operating functions: heating, cooling, and delayed high-temperature monitoring.

For heating, the PLC compares N7:0 against 70 using a LES instruction. The Heater operates only when the temperature is below this value.

For cooling, a separate GRT instruction compares N7:0 against 80. The Cooling Fan operates only when the temperature is above 80. Using two separate thresholds creates a region between 70°F and 80°F where neither output is active.

When N7:0 exceeds 90, T4:0 begins timing. The Alarm is enabled only after the abnormal temperature condition remains continuously present until the timer reaches its 10-second preset.

7. Engineering Design Decisions

Design Decision	Reason
Used N7:0 for temperature	Provided a simple integer value for simulating process temperature
Used LES N7:0 70	Established a clear lower threshold for heating
Used GRT N7:0 80	Established a separate upper threshold for cooling
Used separate heating and cooling thresholds	Created a 70-80°F region where neither output operates
Used GRT N7:0 90 for alarm monitoring	Separated abnormal high temperature from normal cooling operation
Added TON T4:0	Prevented a brief excursion above 90°F from immediately activating the alarm
Used T4:0/DN for Alarm O:2/2	Made alarm operation dependent on completion of the delay
Kept the alarm non-latching	Version 1 required indication rather than a latched fault or shutdown

One limitation of Version 1 is that the temperature is manually represented through N7:0. The project demonstrates the control decision logic rather than a complete temperature-measurement system.

8. Ladder Logic Description

The Version 1 program contains four functional rungs. Each rung is presented in the approved single-line portfolio format.

8.1 Rung 0000 - Heater Control

```
|----[ LES N7:0 70 ]------( OTE O:2/0 )----|
      Temperature < 70                Heater
```

When N7:0 < 70, Heater O:2/0 energizes. At exactly 70°F, the LES condition is false and the Heater is OFF.

8.2 Rung 0001 - Cooling Fan Control

```
|----[ GRT N7:0 80 ]------( OTE O:2/1 )----|
      Temperature > 80                Cooling_Fan
```

When N7:0 > 80, Cooling Fan O:2/1 energizes. At exactly 80°F, the GRT condition is false.

Temperature	Heater	Cooling Fan
< 70°F	ON	OFF
70-80°F	OFF	OFF
> 80°F	OFF	ON

8.3 Rung 0002 - High-Temperature Alarm Delay

```
|----[ GRT N7:0 90 ]-----[ TON T4:0 ]----|
      Temperature > 90                Alarm_Timer  PRE: 1000  Time Base: 0.01
```

When N7:0 exceeds 90, T4:0 begins accumulating. The temperature must remain continuously above 90°F for the timer to reach its preset. If it returns to 90°F or below before completion, the TON resets.

8.4 Rung 0003 - High-Temperature Alarm

```
|----[ XIC T4:0/DN ]------( OTE O:2/2 )----|
      Alarm_Timer_Done                Alarm
```

When T4:0 reaches its preset, T4:0/DN becomes true and Alarm O:2/2 energizes. The alarm is not latched.

8.5 Actual RSLogix Implementation

The figure below shows the corresponding ladder logic as implemented in RSLogix Micro Starter Lite.

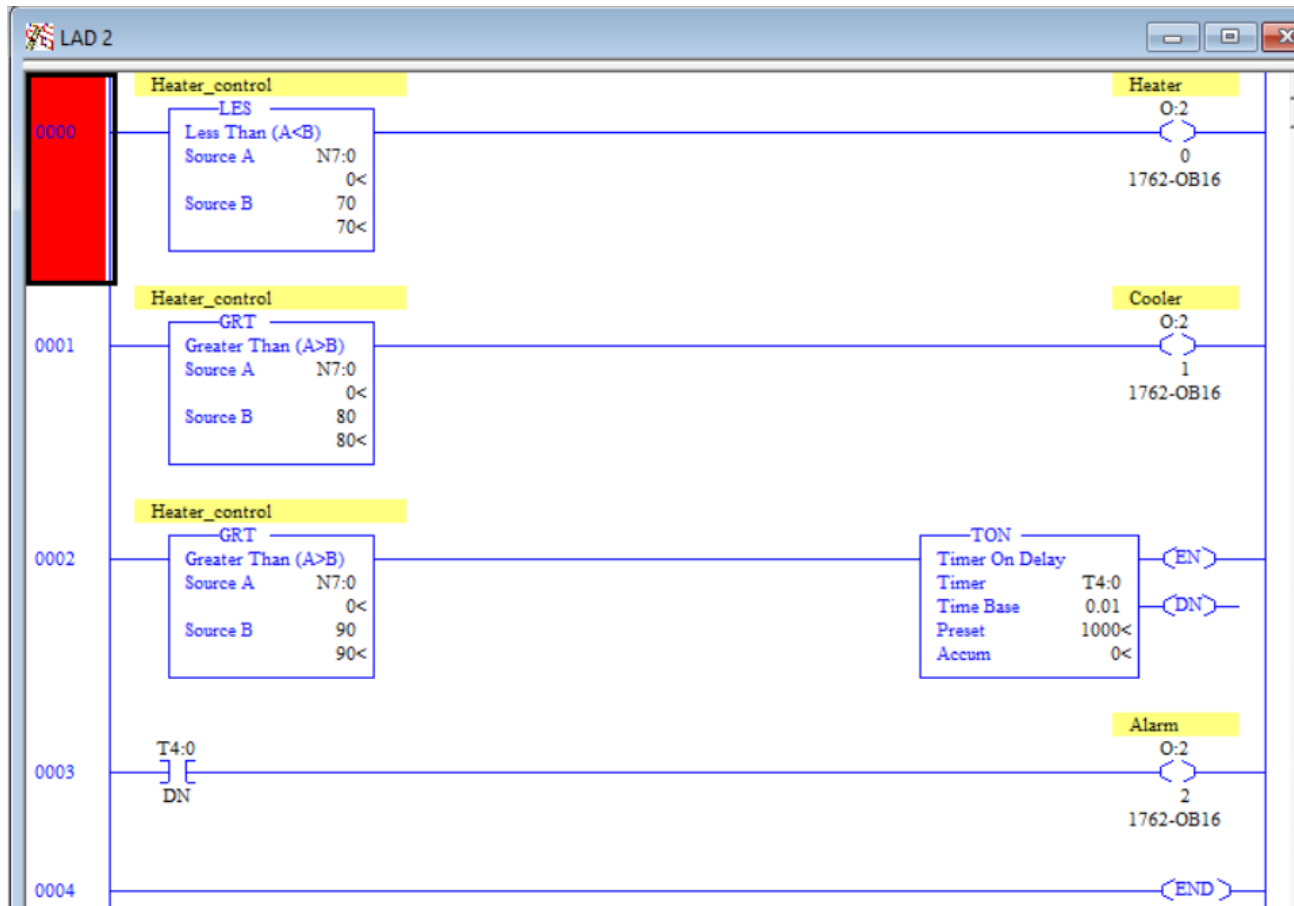


Figure P4.1 - RSLogix Implementation of the Version 1 Temperature Control System

9. Sequence of Operation

- Step 1 - Temperature Value Established:** A simulated temperature value is entered into N7:0.
- Step 2A - Temperature Below 70°F:** The LES comparison becomes true and Heater O:2/0 energizes.
- Step 2B - Temperature from 70°F through 80°F:** Both Heater and Cooling Fan remain OFF.
- Step 2C - Temperature Above 80°F:** Cooling Fan O:2/1 energizes.
- Step 3 - High Temperature Detected:** If temperature rises above 90°F, the Cooling Fan remains ON and T4:0 begins timing.
- Step 4A - Temperature Returns to 90°F or Below Before 10 Seconds:** The timer resets and the Alarm remains OFF.
- Step 4B - Temperature Remains Above 90°F Through Timer Period:** T4:0 reaches its preset and T4:0/DN becomes true.
- Step 5 - Alarm Indication:** Alarm O:2/2 energizes.
- Step 6 - High-Temperature Condition Clears:** T4:0 resets, Done clears, and Alarm turns OFF.

10. Testing & Validation

I tested the program across the heating, deadband, cooling, and delayed alarm operating regions. All seven defined functional tests passed.

Test ID	Test Condition	Expected Result	Observed Result	Status
T-01	N7:0 = 60°F	Heater ON, Fan OFF, Alarm OFF	Heater ON, Fan OFF, Alarm OFF	PASS
T-02	N7:0 = 70°F	Heater OFF, Fan OFF, Alarm OFF	Heater OFF, Fan OFF, Alarm OFF	PASS
T-03	N7:0 = 75°F	Heater OFF, Fan OFF, Alarm OFF	Heater OFF, Fan OFF, Alarm OFF	PASS
T-04	N7:0 = 80°F	Heater OFF, Fan OFF, Alarm OFF	Heater OFF, Fan OFF, Alarm OFF	PASS
T-05	N7:0 = 85°F	Heater OFF, Fan ON, Alarm OFF	Heater OFF, Fan ON, Alarm OFF	PASS
T-06	N7:0 = 95°F for <10 s	Heater OFF, Fan ON, Alarm OFF	Heater OFF, Fan ON, Alarm OFF	PASS
T-07	N7:0 = 95°F until 10 s	Heater OFF, Fan ON, Alarm ON	Heater OFF, Fan ON, Alarm ON	PASS

Validation Summary

Testing at exactly 70°F and 80°F confirmed the strict comparison boundaries. The 75°F test confirmed the deadband, while 85°F confirmed cooling operation. The two 95°F tests verified both sides of the alarm delay. All seven Version 1 functional tests passed.

The testing represents functional PLC program validation within the project development environment, not industrial commissioning or safety validation.

11. Results

The completed Temperature Control System successfully performed the required heating, cooling, deadband, and delayed alarm functions. The Heater energized below 70°F, the Cooling Fan energized above 80°F, and neither output operated within the 70-80°F deadband.

A brief excursion above 90°F did not immediately produce an alarm. Alarm O:2/2 energized only when the high-temperature condition remained present until the 10-second timer preset was reached. All seven functional tests passed.

12. Challenges & Troubleshooting

A useful troubleshooting path for this project is: Temperature Value -> Comparison Result -> Heater/Cooling Output -> High-Temperature Comparison -> Timer -> Done Bit -> Alarm.

Symptom	Possible Cause	Diagnostic Check	Corrective Action
Heater does not turn ON	Temperature is not below 70 or LES comparison is incorrect	Monitor N7:0 and Rung 0000	Verify temperature value and LES parameters
Heater remains ON unexpectedly	Temperature remains below 70 or output logic is incorrect	Monitor N7:0 and O:2/0	Verify comparison and output address
Cooling Fan does not turn ON	Temperature is not greater than 80 or GRT comparison is incorrect	Monitor N7:0 and Rung 0001	Verify GRT parameters
Both Heater and Fan are OFF	Temperature may be inside intended 70-80°F deadband	Check N7:0	Confirm normal operation

Alarm timer does not run	Temperature is not greater than 90	Monitor N7:0 and T4:0	Verify high-temperature condition
Alarm never turns ON	Timer does not reach preset or Done bit is incorrect	Monitor T4:0.ACC and T4:0/DN	Verify timer configuration and Rung 0003
Alarm remains ON	Temperature remains above 90 or timer has not reset	Monitor N7:0 and T4:0/DN	Clear condition and verify TON reset

A key lesson was the importance of checking comparison boundaries. GRT N7:0 80 means greater than 80, not greater than or equal to 80; therefore, a value of exactly 80 correctly leaves the Cooling Fan OFF.

13. Lessons Learned

This project helped me understand how numeric process values can be used directly in PLC decision logic.

I learned the practical difference between LES and GRT instructions and the importance of checking exact threshold values during testing.

The separate heating and cooling thresholds create a region where neither output operates rather than switching directly between heating and cooling at one value.

The project also demonstrated how a timer can qualify an alarm condition so that a short-duration excursion does not immediately activate an alarm.

14. Industrial Relevance

Temperature monitoring and control are common in HVAC systems, industrial ovens, plastic molding machines, heat-treatment furnaces, chemical reactors, and food-processing systems.

A real industrial application would normally require actual temperature instrumentation, analog input scaling, sensor failure detection, adjustable setpoints, output interlocks, alarm management, and possibly PID control depending on the process.

This Version 1 project therefore represents a foundation temperature-control exercise rather than a production-ready temperature-control or safety system.

15. Transition to Industrial Version (Version 2)

Version 2 can retain the basic requirements while introducing a more realistic industrial control architecture.

Version 2 Improvement	Engineering Benefit
Analog temperature input	Replaces manually entered N7:0 with process measurement
Raw-to-engineering-unit scaling	Converts analog signal into meaningful temperature units
Sensor range validation	Detects invalid or failed temperature signals
Operator-adjustable setpoints	Allows authorized threshold changes without editing ladder logic
HMI temperature display	Provides real-time process visibility
Auto/Manual modes	Supports automatic operation and maintenance control
Heater/Cooler permissives	Prevents outputs unless required conditions are satisfied
Mutual-exclusion logic	Explicitly prevents conflicting heating and cooling commands
High-temperature warning	Provides early abnormal-condition indication
Latched high-temperature fault	Preserves an abnormal condition until deliberate reset
Alarm acknowledgement/reset	Provides controlled operator response
Output/fault feedback	Improves troubleshooting and verifies equipment response
Trend display	Allows temperature behavior to be observed over time
PID control where justified	Provides proportional regulation when tighter control is required

A stronger architecture could follow: Temperature Sensor -> Analog Scaling -> Signal Validation -> Temperature Value -> Heating/Cooling Decision -> Permissive & Interlock Checks -> Output Commands -> High-Temperature Monitoring -> Alarm/Fault Handling -> HMI Status & Diagnostics.

PID control should only be added if the Version 2 process objective requires continuous proportional regulation rather than the existing ON/OFF strategy.

16. Project Summary

I developed the Temperature Control System to build practical experience using numeric comparison instructions and timer-based alarm logic. The system used N7:0 to control Heater O:2/0 below 70°F, Cooling Fan O:2/1 above 80°F, and a neutral region between the thresholds.

For high-temperature monitoring, T4:0 was enabled when the temperature exceeded 90°F. Alarm O:2/2 energized only when the condition remained present until the timer reached its 10-second preset.

All seven defined functional tests passed, including exact 70°F and 80°F boundary tests and tests on both sides of the high-temperature alarm delay.

This project strengthened my understanding of integer data, comparison instructions, operating thresholds, deadband behavior, timers, and alarm qualification. It provides a clear foundation for Version 2 with real analog measurement, signal scaling, adjustable setpoints, HMI integration, diagnostics, and fault handling.